Explicit nonpolytopal Bier spheres

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Abstract. We construct explicit Bier spheres that admit no convex realization, answering a question of Holleben and Yang. One example has 290 365 vertices and dimension 145 181, and its defining complex has dimension nine. The obstruction converts a hypothetical convex realization into a matrix of bounded sign rank. A rectangular submatrix of an explicit Hadamard matrix violates this bound and specifies the required face memberships by finite-field formulas. The same obstruction gives examples with no ghost labels, with combinatorial central symmetry, and with a defining complex of dimension three.

1. Introduction

Bier's construction associates a simplicial sphere to every nonempty proper simplicial complex. Counting shows that most of these spheres are nonpolytopal [1], but does not specify an example. Holleben and Yang [3] proved that every Bier sphere with at most twelve vertices is polytopal and asked for an explicit nonpolytopal sphere. We construct one by encoding a matrix whose sign rank exceeds a bound imposed by every convex realization.

Let K be a nonempty proper simplicial complex on a labelled ground set V. Singleton labels need not be faces; an absent singleton is called a *ghost*. The Alexander dual and Bier sphere are

$$K^* = \{B \subseteq V : V \setminus B \notin K\}, \qquad \mathrm{Bier}(K) = \{A \cup B' : A \in K,\ B \in K^*,\ A \cap B = \varnothing\},$$

where B' is a copy of B on a disjoint primed ground set. The deleted join $\mathrm{Bier}(K)$ is a sphere of dimension $|V| - 2$ [1].

Theorem 1. *There is an explicitly specified complex of dimension nine whose Bier sphere is nonpolytopal, has* 290 365 *actual vertices, and has dimension* 145 181. *There are also nonpolytopal examples with all ground labels present, with a self-dual input complex, and with an input complex of dimension three.*

The obstruction depends on a small core of labels. An additional label records whether each selected core face extends to a face with that label. These membership signs form a matrix. Convexity forces its sign rank to be bounded solely in terms of the number of core labels, independently of the number of additional labels.

For a sign matrix $H \in \{-1, +1\}^{r \times s}$, define

$$\mathrm{signrank}(H) = \min\{\operatorname{rank} M : H_{it} M_{it} > 0 \text{ for all } i, t\}.$$

The precise geometric obstruction is as follows.

Theorem 2. *Let $\mathcal{K}$ be a proper complex on $E = U \sqcup \{a_1, \dots, a_s\}$, containing every singleton, where $|U| = m$. Choose faces $S_1, \dots, S_r \subseteq U$ of one fixed positive cardinality, and define*

$$H_{it} = +1 \quad \Longleftrightarrow \quad S_i \cup \{a_t\} \in \mathcal{K}.$$

Regard $\mathcal{K}$ as a complex Δ on $E \sqcup \{g\}$, with g a ghost. If $\mathrm{Bier}(\Delta)$ *is polytopal, then*

$$\mathrm{signrank}(H) \leq m^2 - m + 1.$$

For selected faces of arbitrary cardinalities, the bound $m^2 - m + 2$ holds instead.

Supporting hyperplanes express a matrix with signs H through Schur complements of one matrix. The required principal inverses, padded by zeros, satisfy linear identities leaving at most $m^2 - m + 1$ dimensions. This proves the bound using only the selected memberships; the other faces of the complex are unrestricted.

For the principal example, take $m = 20$ and retain $s = 381^2 + 1$ columns of a Hadamard matrix of order $147\,456 = 384^2$. Forster's spectral bound [2] and Theorem 2 would give

$$381 < \sqrt{s} \leq \mathrm{signrank}(H) \leq 20^2 - 20 + 1 = 381.$$

Section 3 specifies the matrix and the face set by elementary finite-field formulas. The final constructions preserve the obstruction while changing the ghost, symmetry, or input-dimension conditions.

2. **The sign-rank obstruction**

Put $q = |E|$. All q unprimed labels in E and all $q + 1$ primed labels occur in $\mathrm{Bier}(\Delta)$: the complements defining the primed singletons either contain the ghost or equal the nonface E. Thus the sphere has $2q + 1$ actual vertices and dimension $q - 1$.

In a hypothetical realization, denote the labelled vertices by $x_i = i$, $p_i = i'$ for $i \in E$, and $z = g'$. The facets we need are

$$\begin{aligned} T_A &= \{x_i : i \in A\} \cup \{p_i : i \in E \setminus A\} && (A \in \mathcal{K}), \\ F_{B,a} &= \{z\} \cup \{x_i : i \in B\} \cup \{p_i : i \in E \setminus (B \cup \{a\})\} && (B \in \mathcal{K},\ B \cup \{a\} \notin \mathcal{K}). \end{aligned}$$

In each case the positive set is a face, the complement of the primed set is a nonface, and there are q vertices. These are therefore facets by the definition of the Bier sphere.

The two kinds of facets distinguish an extension from a nonextension. If $S \cup \{a\}$ is a face, replacing p_a by x_a in T_S gives another facet. If it is a nonface, the

corresponding facet contains z and omits both p_a and x_a. After using z as the origin, these alternatives determine opposite signs for one coordinate of x_a in the basis supplied by T_S. The next lemma computes that coordinate.

Lemma 3. *If* $\mathrm{Bier}(\Delta)$ *is polytopal, there are a real* $m \times m$ *matrix* G*, row vectors* u_t*, column vectors* v_t*, and scalars* d_t *such that every* $G[S_i, S_i]$ *is invertible and*

$$H_{it}\big(d_t - u_t[S_i]G[S_i,S_i]^{-1}v_t[S_i]\big) > 0.$$

Proof. The simplex facet $T_\varnothing$ and the vertex z outside its affine hull form an affine basis. Normalize the realization to

$$z = 0, \qquad p_i = e_i, \qquad x_i = -N_i,$$

where N_i is column i of a real $q \times q$ matrix N. For a selected face S, let C_S have columns x_j for $j \in S$ and e_j otherwise. These columns form the simplex facet T_S outside the origin, so C_S is invertible. Expanding in its standard basis columns shows that $N[S,S]$ is invertible as well.

For an anchor $a = a_t$, write $\alpha = C_S^{-1}x_a$. Solving first in the rows S and then in row a gives

$$\alpha_a = -N_{aa} + N[a,S]N[S,S]^{-1}N[S,a].$$

We claim that $-\alpha_a$ has the prescribed sign H_{it}.

If $S \cup \{a\}$ is a face, normalize the supporting functionals of T_S and $T_{S\cup\{a\}}$ to $h \le 1$ and $h' \le 1$. Every column of C_S has h-value one, while all except column a also have h'-value one. Since x_a lies outside the first facet and p_a outside the second,

$$\sum_j \alpha_j = h(x_a) < 1, \qquad 1 - \sum_j \alpha_j = \alpha_a\big(h'(p_a) - 1\big).$$

The left side is positive and the parenthesized factor negative, so $\alpha_a < 0$.

If $S \cup \{a\}$ is a nonface, use the facet $F_{S,a}$. Its supporting linear functional ℓ may be chosen zero on the facet and positive at every other vertex. It vanishes on all columns of C_S except column a, whence

$$0 < \ell(x_a) = \alpha_a \ell(p_a), \qquad \ell(p_a) > 0.$$

Thus $\alpha_a > 0$. The claim follows with

$$G = N[U,U], \quad u_t = N[a_t,U], \quad v_t = N[U,a_t], \quad d_t = N_{a_t a_t}. \qquad \square$$

The Schur complement still appears to depend on all entries of a principal inverse. The saving comes from padding each inverse to the same matrix size: multiplication by G recovers the indicator of its index set on the diagonal, on either side. The resulting commutator identities are common linear constraints on every such inverse.

Lemma 4. *Let G be a real $m \times m$ matrix and let S range over subsets on which $G[S,S]$ is invertible. Pad each inverse by zeros to obtain an $m \times m$ matrix W_S, with $W_\varnothing = 0$. Then*

$$\dim \operatorname{span}\{W_S\} \le m^2 - m + 1.$$

Proof. Direct multiplication gives

$$\operatorname{diag}(GW_S) = \operatorname{diag}(W_S G) = \mathbf{1}_S, \qquad \operatorname{tr}(GW_S) = |S|.$$

Join i and j when G_{ij} or G_{ji} is nonzero, and let $m_1, \dots, m_c$ be the component sizes of this support graph. All W_S belong to the block-diagonal matrix space of dimension $\sum_j m_j^2$. For every diagonal matrix D, they satisfy

$$\operatorname{tr}([D,G]W_S) = 0.$$

The map $D \mapsto [D,G]$ has kernel the diagonals constant on each support component, and hence rank $m - c$. The trace pairing on the block-diagonal space is nondegenerate, so these give $m - c$ independent constraints. Therefore

$$\dim \operatorname{span}\{W_S\} \le \sum_j m_j^2 - m + c \le m^2 - m + 1,$$

where the last inequality follows from $m^2 - \sum_j m_j^2 = 2\sum_{i<j} m_i m_j \ge c - 1$. □

Proof of Theorem 2. Let $k > 0$ be the common cardinality of the selected faces. Lemma 3 supplies a matrix with signs H and entries

$$R_{S,t} = d_t - u_t W_S v_t.$$

Because $\operatorname{tr}(GW_S) = k$, its columns are evaluations of the linear functionals

$$W \longmapsto \frac{d_t}{k} \operatorname{tr}(GW) - u_t W v_t.$$

Lemma 4 bounds its rank by $m^2 - m + 1$. Indeed, after choosing a basis of the span of the W_S, the matrix R factors as the matrix of their coordinates times the matrix of these linear functionals. Fixed cardinality is used precisely to express the constant term d_t as a linear function of W_S. Without the common-cardinality assumption, adjoining the constant feature to this matrix space gives $m^2 - m + 2$. □

3. **An explicit Hadamard example**

To encode any sign matrix $H \in \{-1,+1\}^{r\times s}$, choose distinct k-subsets $S_1,\dots,S_r$ of $U=[m]$, with $1\le k<m$ and $s\ge 2$. On $U\sqcup\{a_1,\dots,a_s\}\sqcup\{g\}$, define Δ_H by the following faces:

$$\begin{aligned} &A\subseteq U, && |A|\le k;\\ &\{a_t\}\cup A, && A\subseteq U,\ |A|\le k-1;\\ &\{a_t\}\cup S_i, && H_{it}=+1. \end{aligned} \tag{1}$$

There are no other faces. This is a complex because every proper subset of a face in the third line occurs in the first two. The label g is a ghost, and no face contains two anchors.

Forster's spectral bound [2, Theorem 2.2] states that

$$\operatorname{signrank}(H)\ge\frac{\sqrt{rs}}{\|H\|}\qquad\bigl(H\in\{-1,+1\}^{r\times s}\bigr),$$

where $\|H\|$ is the Euclidean operator norm. In particular, retaining s columns of a Hadamard matrix of order L gives sign rank at least $\sqrt{s}$, since its columns are orthogonal and have norm $\sqrt{L}$. We therefore need

$$(m^2-m+1)^2<s\le L\le\binom{m}{k}.$$

The first inequality defeats the geometric bound; the last leaves enough core faces to encode all rows. We use $m=20$, $k=9$ and an order $L=384^2$, just above the required threshold 381^2. We construct the required Hadamard matrix from a quadratic-character matrix and a binary character matrix.

Let $I=\{\infty\}\sqcup\mathbb{F}_{11}$, ordered by $\infty,0,\dots,10$, and let χ be the quadratic character of $\mathbb{F}_{11}$, with $\chi(0)=0$. Define

$$K_{\infty,j}=K_{i,\infty}=1,\qquad K_{ii}=-1\ (i\in\mathbb{F}_{11}),\qquad K_{ij}=\chi(j-i)\ (i\ne j\text{ in }\mathbb{F}_{11}).$$

This specifies a 12×12 sign matrix. If $Q_{ij}=\chi(j-i)$, including its zero diagonal, then

$$Q\mathbf{1}=0,\qquad Q^T=-Q,\qquad QQ^T=11I-J.$$

The last identity is the character sum $\sum_t\chi(t)\chi(t-a)=-1$ for $a\ne 0$. One direct proof reduces to $a=1$ and counts the solutions of $v^2=u(u-1)$: under $X=2u-1$, $Y=2v$, this becomes $(X-Y)(X+Y)=1$, with ten solutions. Consequently

$$K=\begin{pmatrix}1 & \mathbf{1}^T\\ \mathbf{1} & Q-I\end{pmatrix},\qquad KK^T=12I.$$

Let $W_{xy} = (-1)^{x\cdot y}$ for $x, y \in \mathbb{F}_2^{10}$, where the dot product is over $\mathbb{F}_2$. Character orthogonality gives $WW^T = 2^{10} I$. Thus

$$\widehat{H} = K \otimes K \otimes W, \qquad \widehat{H}\widehat{H}^T = L I_L, \qquad L = 12^2\, 2^{10} = 147\,456 = 384^2.$$

Index rows and columns of $\widehat{H}$ lexicographically by $I \times I \times \mathbb{F}_2^{10}$, ordering binary vectors by their integer values, and let H consist of its first $s = 381^2 + 1 = 145\,162$ columns. Take $m = 20$, $k = 9$, and assign its rows to the first L nine-subsets of $[20]$ in lexicographic order. This is possible because $\binom{20}{9} = 167\,960 > L$. Formula (1) now defines the principal example completely.

Principal assertion of Theorem 1. Forster's bound gives $\operatorname{signrank}(H) \geq \sqrt{s} > 381$, whereas polytopality would give $\operatorname{signrank}(H) \leq 381$ by Theorem 2. The input has a face of size ten and none larger, so its dimension is nine. There are $20 + s + 1 = 145\,183$ ground labels. Exactly the unprimed ghost is absent from the Bier sphere, giving $2(20 + s) + 1 = 290\,365$ vertices and dimension $20 + s - 1 = 145\,181$. □

4. **Variants of the construction**

4.1. **Removing the ghost**

The ghost made the coordinate normalization convenient, but the nonpolytopality survives making it a vertex. The resulting change to the sphere is a single facet subdivision, which can be reversed in any convex realization.

For the principal example, put $\Delta^+ = \Delta \cup \{\{g\}\}$. Then $\operatorname{Bier}(\Delta^+)$ is the stellar subdivision of $\operatorname{Bier}(\Delta)$ at the all-primed facet $T_\varnothing$. That facet is replaced by

$$\{g\} \cup \{i' : i \in E \setminus \{j\}\}, \qquad j \in E,$$

and all other facets remain unchanged.

A stellar subdivision of a facet can be polytopal only if the original sphere was polytopal. Indeed, in a simplicial d-polytope realizing the subdivision, the new vertex v has exactly d neighbors $u_1, \ldots, u_d$. Its tangent cone has the independent edge rays $u_i - v$, so every point of the polytope has coordinates

$$v + \sum_i \lambda_i (u_i - v), \qquad \lambda_i \geq 0.$$

Every other vertex except these neighbors satisfies $\sum_i \lambda_i > 1$, since otherwise it lies in the corner simplex. Deleting v therefore reverses stacking across the hyperplane $\sum_i \lambda_i = 1$ and realizes the original sphere, a contradiction. Thus $\operatorname{Bier}(\Delta^+)$ is nonpolytopal, has 290 366 actual vertices, and has dimension 145 181. Both copies of every ground label occur.

4.2. **A self-dual input**

Nonpolytopality also survives imposing duality symmetry. Here the original sphere will occur as a vertex link, so a convex realization of the larger sphere would give one of the original.

Let $\mathcal{K}$ be the principal input on E before adjoining the ghost. Add two labels a, b and set

$$\mathcal{L} = 2^E \cup (\{a\} * \mathcal{K}) \cup (\{b\} * \mathcal{K}^*),$$

where the dual is relative to E and no face contains both a, b. Checking whether a subset contains neither, one, or both new labels gives $\mathcal{L}^* = \mathcal{L}$. Moreover,

$$\mathrm{lk}_a \,\mathrm{Bier}(\mathcal{L}) \cong \mathrm{Bier}(\Delta).$$

To verify the link identity, its positive part is a face of $\mathcal{K}$ and cannot contain b. Its primed part is either any disjoint subset of E, or $B \cup \{b\}$ with $B \in \mathcal{K}^*$. These are exactly the dual conditions for $\mathcal{K}$ with b adjoined as a ghost.

Vertex links in boundaries of convex polytopes are boundaries of vertex figures. Hence $\mathrm{Bier}(\mathcal{L})$ is nonpolytopal. It has 290 368 actual vertices and dimension 145 182. Self-duality makes the exchange of each label with its primed copy a simplicial involution taking every nonempty face to a disjoint face. Thus this example is centrally symmetric as a combinatorial sphere.

4.3. **Input dimension three**

To lower the dimension of the input, we can use fewer core labels and many more anchors. Instead of obtaining large sign rank from nearly orthogonal columns, we include every possible column sign pattern. This forces full row rank by an elementary separation argument.

Take the first 74 triples of $[9]$, and index the anchors by $\varepsilon \in \{-1, +1\}^{73}$. Define

$$H_{1,\varepsilon} = 1, \qquad H_{i,\varepsilon} = \varepsilon_{i-1} \quad (2 \le i \le 74).$$

Every real matrix with these signs has independent rows. For if λ were a nonzero row dependence, choose a column whose signs agree with the nonzero signs of λ, up to one common sign chosen to make the first coordinate positive. Every nonzero summand in that column's dependence has the same sign, a contradiction. Thus $\mathrm{signrank}(H) = 74$.

There are $\binom{9}{3} = 84$ available row sets. Apply (1) with $m = 9$, $k = 3$. Theorem 2 would give $\mathrm{signrank}(H) \le 9^2 - 9 + 1 = 73$ in a convex realization. The resulting nonpolytopal Bier sphere has $2^{74} + 19$ actual vertices and dimension $2^{73} + 8$, while the input has dimension three. This proves the remaining assertion of Theorem 1.

More generally, whenever $\binom{m}{k} \geq m^2 - m + 2$, the same all-sign-columns construction gives a nonpolytopal Bier sphere from an input of dimension k.

References

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